

COMPLETED AND PRELIMINARY FOUNDATIONS



EXPERIMENT-CENTERED AIMS

Measured 4D grain evolution is the scientific target

Local learning is the tool; experimental behavior defines the claims



AIM 1

Establish reliable experimental learning

Cross-window generalization, uncertainty, and applicability limits



AIM 2

Quantify domain-specific surrogate fidelity

Paired reference and trained evolution within each domain



AIM 3

Interpret learned evolution mechanisms

Robust rules tested against independent physical evidence